in detail in section 9.4. The lifetime is how long the DI container should use a given object to fulfill a service's dependencies.

Instead of providing a single instance that the container will always use, you can provide a function that the container invokes when it needs an instance of the type, as shown in figure 9.3.

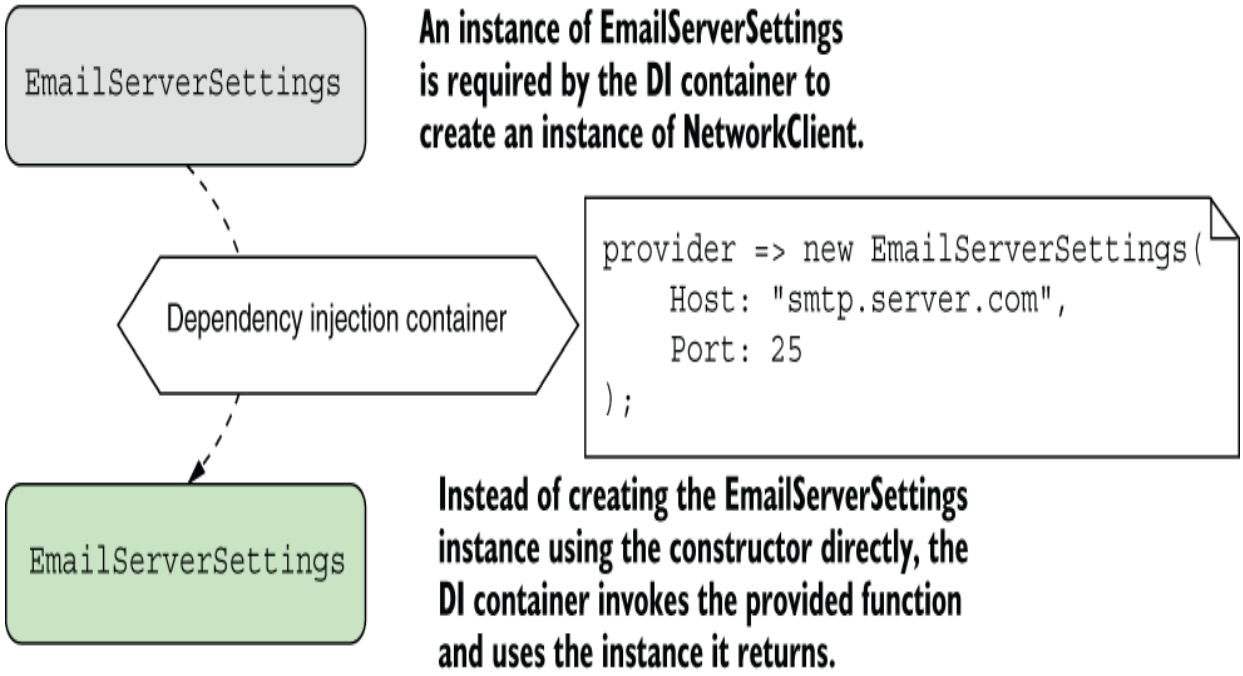


Figure 9.3 You can register a function with the DI container that will be invoked whenever a new instance of a service is required.

NOTE Figure 9.3 is an example of the factory pattern, in which you define how a type is created. Note that the factory functions must be synchronous; you can't create types asynchronously by (for example) using `async`.

The easiest way to register a service using the factory pattern is with a lambda function (an anonymous delegate), in which the container creates a new `EmailServerSettings` object whenever it's needed, as shown in the following listing.